

EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *loop moves, B static*

$$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = - \frac{d\Phi}{dt} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

N-turn coil *transformer, N turns*

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \quad \lambda = N\Phi$$

Uniform B shortcut *B same everywhere on loop*

$$V_{\text{emf}} = - \frac{\partial B}{\partial t} \cdot A$$

Long wire + rect loop *wire near loop, r: a → b, height c*

$$V_{\text{emf}} = - \int_0^c \int_a^b \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz$$

SURFACE ELEMENT ds

$d\mathbf{s} = \hat{n} da$ (RHR gives $\hat{n}$: curl fingers in loop direction, thumb = $\hat{n}$)

Loop plane	$d\mathbf{s}$
x-y	$\hat{z} dx dy$ or $\hat{z} r dr d\phi$
y-z	$\hat{x} dy dz$
x-z	$\hat{y} dx dz$
coplanar / toroid	$\hat{\phi} dr dz$

Uniform B: $\iint da = A$ (just the loop area)

DOT PRODUCTS $\hat{a} \cdot \hat{b}$

Same direction = 1; perpendicular = 0

	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$
$\hat{x}$	1	0	0	0	0
$\hat{y}$	0	1	0	0	0
$\hat{z}$	0	0	1	0	0
$\hat{\phi}$	0	0	0	0	1

CROSS PRODUCT $\mathbf{A} \times \mathbf{B}$ (determinant)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_y B_z - A_z B_y) \hat{x} - (A_x B_z - A_z B_x) \hat{y} + (A_x B_y - A_y B_x) \hat{z}$$

Also used for curl:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z row survives.

FIELDS

$\mathbf{E}$ — electric field (V/m)
 $\mathbf{B}$ — mag. flux density (T)
 $\mathbf{H}$ — mag. field intensity (A/m)
 ρ_v — charge density (C/m³)
 ρ_{v0} — initial charge density
 $\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$ — flux (Wb)
 $\lambda = N\Phi$ — flux linkage (Wb)

CURRENTS

$J_c = \sigma E$ — cond. density (A/m²)
 $J_d = \varepsilon \partial E / \partial t$ — disp. density
 $I_c = J_c \cdot A$ — cond. current (A)
 $I_d = J_d \cdot A$ — disp. current (A)
 $R = d/(\sigma A)$ — resistance (Ω)
 $C = \varepsilon_r \varepsilon_0 A/d$ — capacitance (F)
 $\tau_r = \varepsilon_r \varepsilon_0 / \sigma$ — relax. time (s)

MATERIALS

$\mu_0 = 4\pi \times 10^{-7}$ H/m
 μ_r — relative permeability
 $\mu = \mu_r \mu_0$ — permeability
 $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m
 ε_r — relative permittivity
 $\varepsilon = \varepsilon_r \varepsilon_0$ — permittivity
 σ — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
 R — resistance (Ω)
 C — capacitance (F)
 $\omega = 2\pi f$ — angular freq (rad/s)
 $k = \omega \sqrt{\mu \varepsilon}$ — wave number (m⁻¹)
 $\lambda = 2\pi/k$ — wavelength (m)
 $c = 3 \times 10^8$ m/s — speed of light

PHASOR & GEOMETRY

$\eta = \sqrt{\mu/\varepsilon}$ — intrinsic imp. (Ω)
 $\eta_0 = 120\pi \approx 377 \Omega$ (free space)
 $k/(\omega \mu) = 1/\eta$
 j — imaginary unit ($j^2 = -1$)
 $\tilde{\mathbf{E}}$ — phasor (drop $e^{j\omega t}$)
 A — area (m²); d — sep. (m)
 N — turns; r — radial dist. (m)

B FIELDS

Infinite wire *single wire carrying I, N = 1*

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}$$

Inside toroid *coil of N turns carrying I*

$$\mathbf{B} = \hat{\phi} \frac{\mu N I}{2\pi r} \quad \mu = \mu_r \mu_0$$

N in B only if the source (current-carrying thing) is a multi-turn coil. Single wire $\Rightarrow N = 1$, drop it. N in $V_{\text{emf}} = -Nd\Phi/dt$ is the receiver coil turns.

$\partial B/\partial t$ TABLE

$B(t)$	$\partial B/\partial t$
$B_0 \sin(\omega t)$	$B_0 \omega \cos(\omega t)$
$B_0 \cos(\omega t)$	$-B_0 \omega \sin(\omega t)$
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
B_0 (const)	0

SIGN → CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	$- \rightarrow +$ terminal	$-\hat{\phi}$ (CW)
> 0	$+ \rightarrow -$ terminal	$+\hat{\phi}$ (CCW)
$= 0$	—	none

Lenz: induced I always opposes the change in Φ .

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a$ to $r = b$

$$\int_a^b \frac{dr}{r} = \ln \frac{b}{a} = \ln b - \ln a$$

$$\ln \frac{b}{a} = \ln b - \ln a$$

$$\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182$$

ROTATING LOOP — GENERATOR

$A = \text{loop area (m}^2\text{)}$

$$\Phi = AB_0 \cos(\omega t + \phi_0)$$

$$V_{\text{emf}} = AB_0 \omega \sin(\omega t + \phi_0)$$

$$I_{\text{amp}} = \frac{AB_0 \omega}{R} \quad \omega = 2\pi f = 2\pi \frac{\text{rpm}}{60}$$

SI PREFIXES

Prefix	Sym	Value	Prefix	Sym	Value
pico	p	10^{-12}	kilo	k	10^3
nano	n	10^{-9}	mega	M	10^6
micro	μ	10^{-6}	giga	G	10^9
milli	m	10^{-3}	tera	T	10^{12}

j IDENTITIES

Expression	Result	Expression	Result
j^2	-1	$j \times (-j)$	1
$1/j$	$-j$	$-1/j$	j
j^3	$-j$	j^4	1
$e^{-j\pi/2}$	$-j$	$e^{j\pi/2}$	j
$e^{j\pi}$	-1	$e^{j2\pi}$	1

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$E = \frac{V(t)}{d} \quad J_c = \sigma E \quad I_c = J_c \cdot A = \frac{\sigma A}{d} V(t)$$

$$I_c = \underbrace{\frac{\sigma A}{d}}_{1/R} V(t) \longleftrightarrow I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}$$

Current proportional to V directly $\rightarrow$ behaves like a resistor.
Displacement *capacitive, changing E*

$$J_d = \varepsilon \frac{\partial E}{\partial t} \quad I_d = J_d \cdot A = \frac{\varepsilon_r \varepsilon_0 A}{d} \frac{dV}{dt}$$

$$I_d = \underbrace{\frac{\varepsilon_r \varepsilon_0 A}{d}}_C \frac{dV}{dt} \longleftrightarrow I = C \frac{dV}{dt} \Rightarrow C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

Current proportional to $dV/dt \rightarrow$ behaves like a capacitor.
Lossy cap circuit *both σ and ε_r present*

$$R \parallel C \text{ across } V(t): \quad I = I_c + I_d = \frac{V}{R} + C \frac{dV}{dt}$$

CHARGE RELAXATION

Decay *charge injected at $t = 0$*

$$\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \quad \tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma}$$

Find σ from decay data

$$\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \varepsilon_r \varepsilon_0}{t_1}$$

Larger τ_r = slower dissipation = better insulator.

PHASOR METHOD ($\mathbf{E} \rightarrow \mathbf{H}$)

- Key relation** $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$
- Euler split** $e^{j(\omega t - kz)} = e^{j\omega t} \cdot e^{-jkz}$ strip $e^{j\omega t}$, keep e^{-jkz}
- Time $\rightarrow$ phasor conversions**
 $\cos(\omega t - kz) = \text{Re}\{e^{j(\omega t - kz)}\} = \text{Re}\{e^{j\omega t} \cdot e^{-jkz}\} \rightarrow e^{-jkz}$
 $\sin(\omega t - kz) = \cos(\omega t - kz - \frac{\pi}{2}) = \text{Re}\{e^{j(\omega t - kz - \pi/2)}\} = \text{Re}\{e^{-j\pi/2} \cdot e^{j\omega t} \cdot e^{-jkz}\} \rightarrow -j e^{-jkz}$
- Faraday in phasor domain** (plane wave: only $\partial/\partial z$ survives)

$$\tilde{\mathbf{H}} = - \frac{1}{j\omega \mu} \nabla \times \tilde{\mathbf{E}}$$

- Phasor $\rightarrow$ time domain** $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$
 $\text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$
 $\text{Re}\{-j e^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

UNIT CIRCLE

	Angle	Deg	cos	sin	Q
0	0	0	1	0	—
$\pi/6$	30	$\sqrt{3}/2$	$1/2$	1	
$\pi/4$	45	$\sqrt{2}/2$	$\sqrt{2}/2$	1	
$\pi/3$	60	$1/2$	$\sqrt{3}/2$	1	
$\pi/2$	90	0	1	—	
$2\pi/3$	120	$-1/2$	$\sqrt{3}/2$	2	
$3\pi/4$	135	$-\sqrt{2}/2$	$\sqrt{2}/2$	2	
$5\pi/6$	150	$-\sqrt{3}/2$	$1/2$	2	
π	180	-1	0	—	
$7\pi/6$	210	$-\sqrt{3}/2$	$-1/2$	3	
$5\pi/4$	225	$-\sqrt{2}/2$	$-\sqrt{2}/2$	3	
$4\pi/3$	240	$-1/2$	$-\sqrt{3}/2$	3	
$3\pi/2$	270	0	-1	—	
$5\pi/3$	300	$1/2$	$-\sqrt{3}/2$	4	
$7\pi/4$	315	$\sqrt{2}/2$	$-\sqrt{2}/2$	4	
$11\pi/6$	330	$\sqrt{3}/2$	$-1/2$	4	
2π	360	1	0	—	